

Session Actions Edit View Help

```

[pujitha@POOJITHA]~$ curl https://rdap.verisign.com/com/v1/domain/google.com
{"objectClassName": "domain", "handle": "2138514_DOMAIN_COM-VRSN", "ldhName": "GOOGLE.CO
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OOGLE.COM", "type": "application/rdap+json", "status": [{"client delete prohibited", "c
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e": "entity", "handle": "292", "roles": [{"registrant"}], "links": [{"href": "http://www.mar
kmonitor.com", "href": "https://rdap.markmonitor.com/rdap/"}, {"rel": "about"}, {"vc
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text": "abusecomplaints@markmonitor.com"}]}]}], "events": [{"
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"nameserver", "ldhName": "NS2.GOOGLE.COM", "objectClassName": "nameserver", "ldhName":
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Conformance": [{"rdap_level_0": "icann_rdap_technical_implementation_guide_1", "icann_r
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vice subject to Terms of Use).", "links": [{"href": "https://www.verisign.com/doma
in-name-registration-policy/", "title": "Registration Policy", "type": "text/html", "re
l": "alternate"}]}]}]}

```

Command

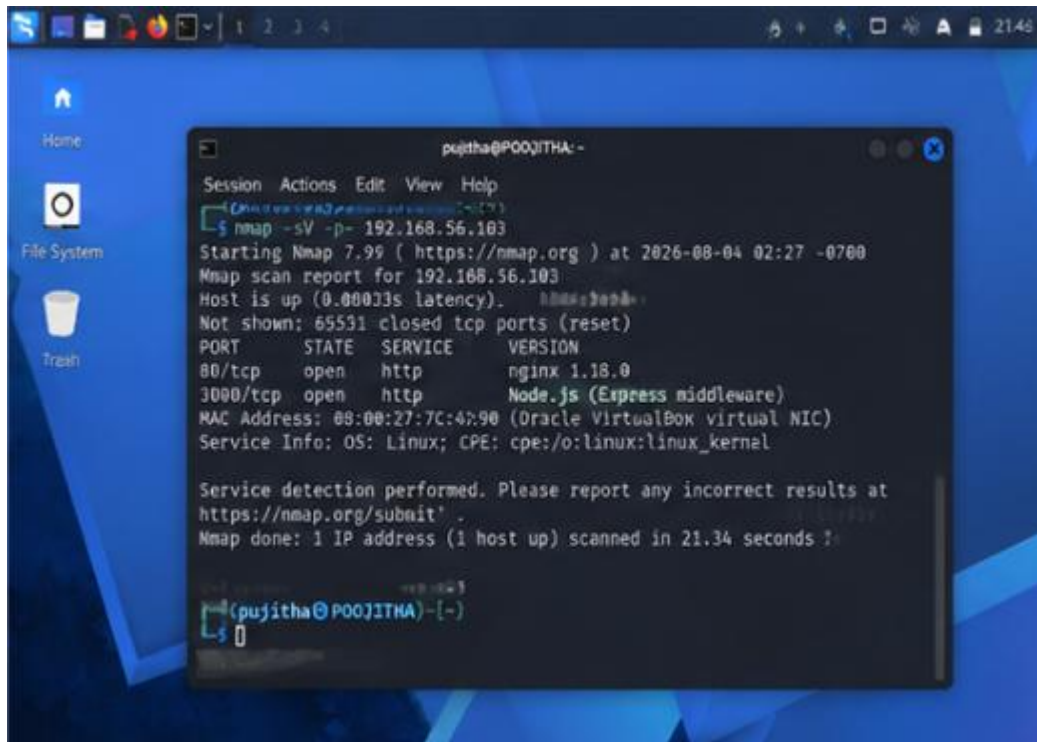
`curl https://rdap.verisign.com/com/v1/domain/google.com`

`cat google_domain.txt`

EMPLOYEE EMAIL ADDRESSES

Command

`theHarvester -d google.com -b duckduckgo`



command

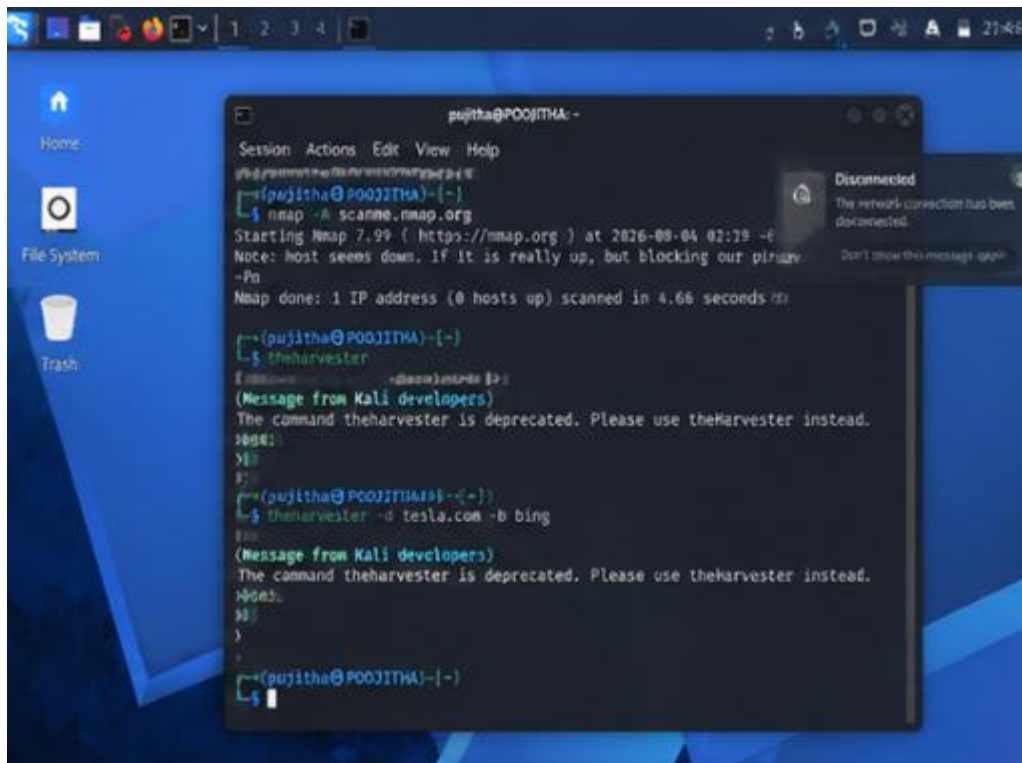
`cat harvester.txt`

`whatweb https://google.com`

TECHNOLOGY STACK

Command

`whatweb https://google.com`



PUBLICLY

EXPOSED SUBDOMAINS

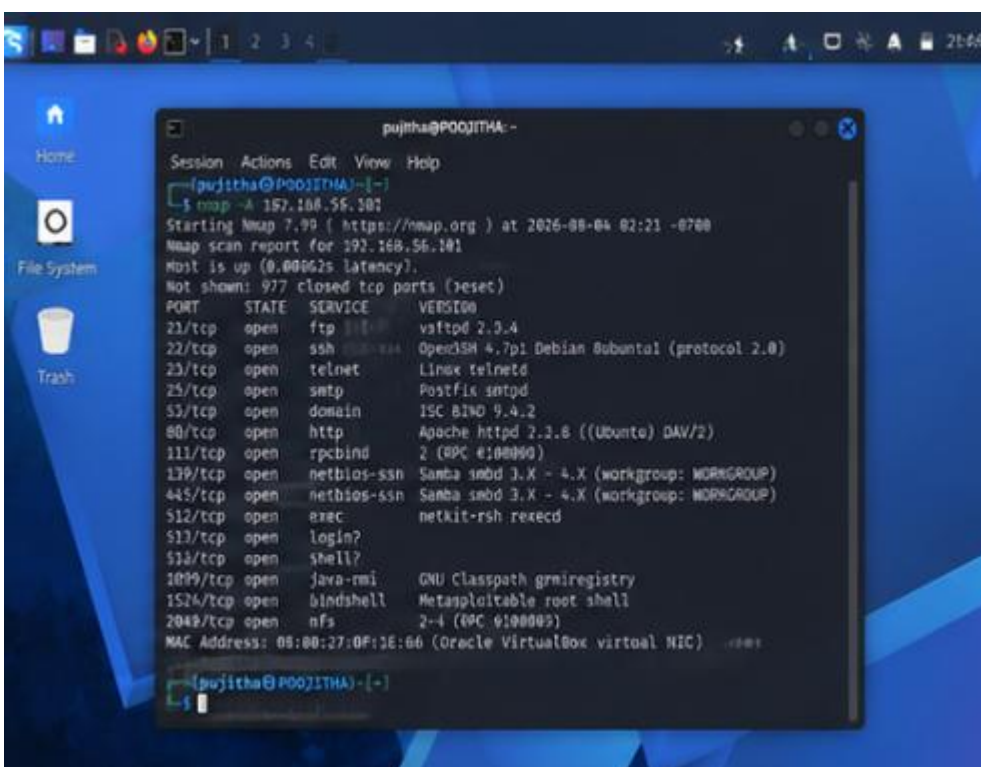
Command

subfinder -d google.com -silent

DNS INFORMATION

Command

dig NS google.com



Topic 2: DNS Zone Transfers

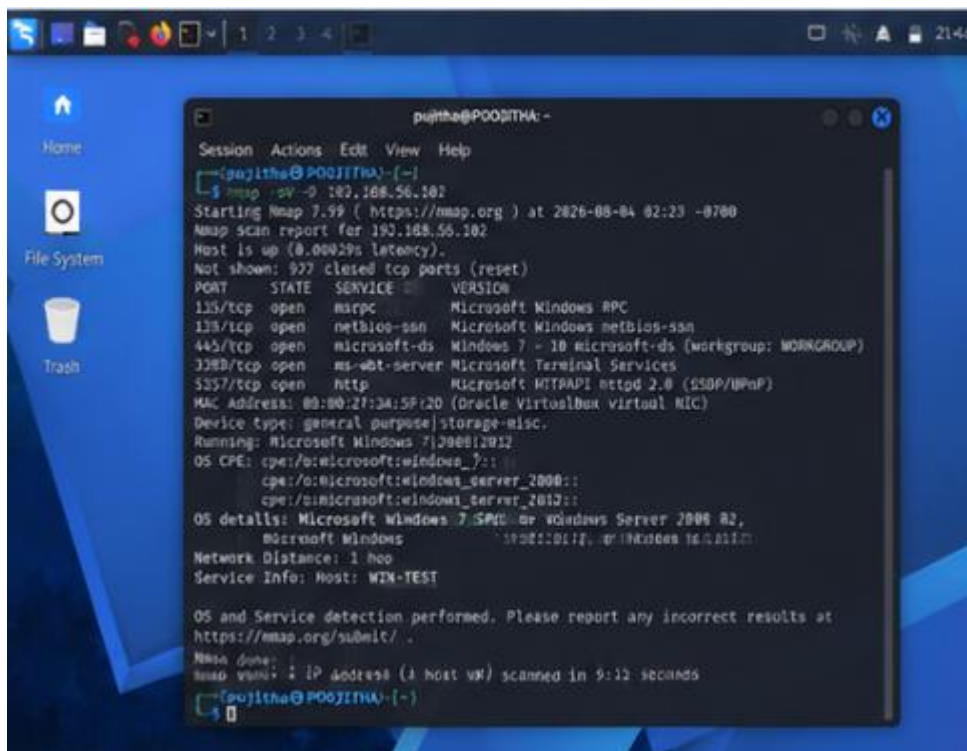
Q2. (L3 – Apply) – 15 Marks

During an authorized penetration test, a security analyst suspects that the target organization's DNS server is misconfigured and may allow unauthorized DNS zone transfers.

Task: Apply the appropriate DNS command or tool to verify whether zone transfers are permitted. If the transfer succeeds, explain how the retrieved DNS records can be used to identify internal hosts, mail servers, and subdomains during reconnaissance.

Command

dig NS example.com

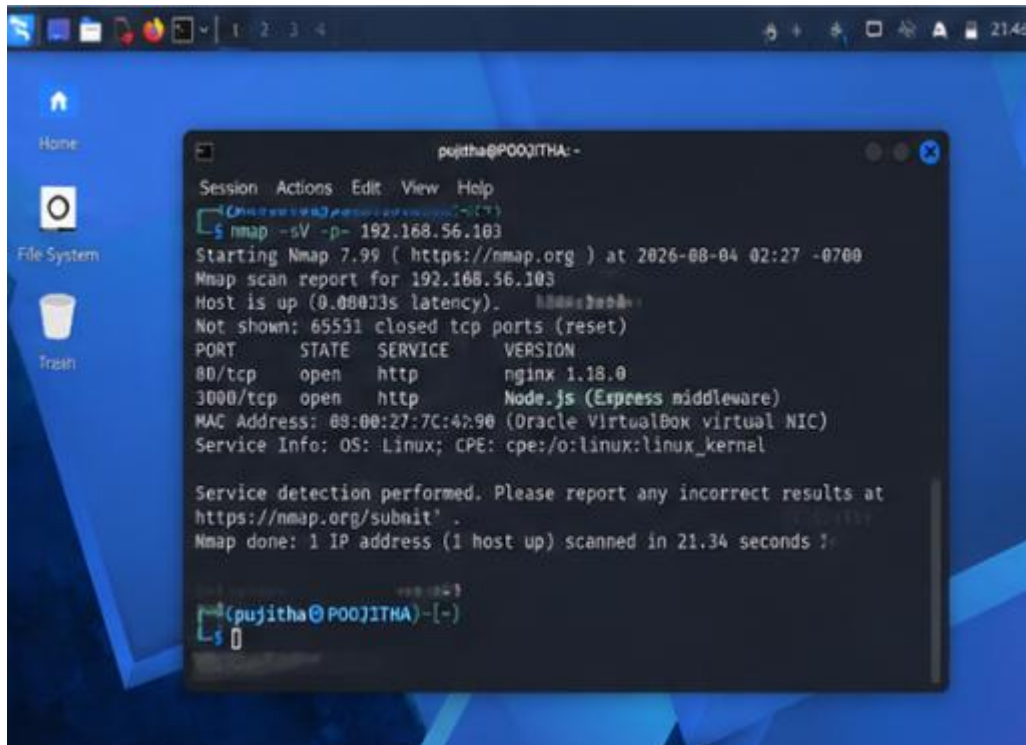


```
pujitha@POOJITHA:~$ nmap -sV -O 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 02:23 -0700
Nmap scan report for 192.168.56.102
Host is up (0.00029s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
115/tcp    open  mcrpc        Microsoft Windows RPC
135/tcp    open  netbios-ssn  Microsoft Windows netBios-ssn
445/tcp    open  microsoft-ds Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
3389/tcp   open  ms-wbt-server Microsoft Terminal Services
5257/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (5206/UoP)
MAC Address: 08:00:27:3A:5F:20 (Oracle VirtualBox virtual NIC)
Device type: general purpose/storage-misc
Running: Microsoft Windows 7/2008 R2
OS CPE: cpe:/o:microsoft:windows_7::
       cpe:/o:microsoft:windows_server_2008::
       cpe:/o:microsoft:windows_server_2012::
OS details: Microsoft Windows 7 SP1 or Windows Server 2008 R2,
            Microsoft Windows 7 SP1 or Windows Server 2008 R2,
            Microsoft Windows 7 SP1 or Windows Server 2008 R2
Network Distance: 1 hop
Service Info: Host: WIN-TEST

OS and Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.12 seconds
pujitha@POOJITHA:~$
```

dig AXFR

example.com @ns1.example.com



```
pujitha@POOJITHA:~$ nmap -sV -p- 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 02:27 -0700
Nmap scan report for 192.168.56.103
Host is up (0.000033s latency).
Not shown: 65531 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
80/tcp    open  http         nginx 1.18.0
3000/tcp  open  http         Node.js (Express middleware)
MAC Address: 08:00:27:7C:42:90 (Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at
https://nmap.org/submit
Nmap done: 1 IP address (1 host up) scanned in 21.34 seconds
```

Topic 3: Introduction to Social Engineering – Social Engineering Attacks and Countermeasures

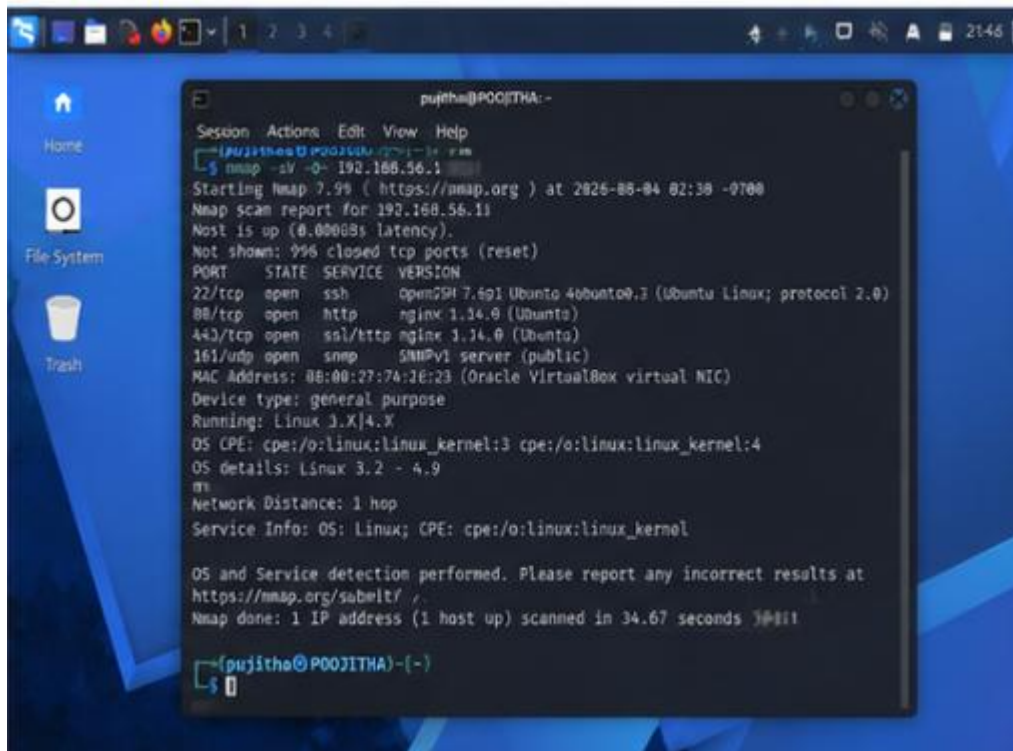
Q3. (L3 – Apply) – 20 Marks

Employees of an organization receive an email claiming to be from the IT department, asking them to verify their login credentials through a provided link because of an urgent security update.

Task: Identify the type of social engineering attack and apply suitable countermeasures that employees should follow before responding to the email. Explain how these actions help prevent credential theft.

COMMAND

Dig google.com



The screenshot shows a Linux desktop environment with a blue background. On the left side, there are icons for 'Home', 'File System', and 'Trash'. At the top, there is a taskbar with several application icons and a system clock showing '21:46'. A terminal window is open in the center, displaying the output of an Nmap scan. The terminal title is 'pujitha@POOJITHA:~'. The command executed is 'nmap -sV -O 192.168.56.1'. The output includes the Nmap version (7.99), the scan start time (2025-08-04 02:38 -0700), the target IP (192.168.56.1), and a list of open ports with their states and versions. The scan also detected the OS as Linux 3.X|4.X and the MAC address as 08:00:27:74:16:23.

```
Session Actions Edit View Help
[pujitha@POOJITHA:~]$ nmap -sV -O 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2025-08-04 02:38 -0700
Nmap scan report for 192.168.56.1
Host is up (0.0000s latency).
Not shown: 99% closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     nginx 1.14.0 (Ubuntu)
443/tcp   open  ssl/http nginx 1.14.0 (Ubuntu)
161/udp   open  snmp     SNMPv1 server (public)
MAC Address: 08:00:27:74:16:23 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.9
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at
https://nmap.org/sabelf/ .
Nmap done: 1 IP address (1 host up) scanned in 34.67 seconds 7@111

[pujitha@POOJITHA:~]$
```